

packages/graph-view/src/utils/elk-layout.ts

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Overview

A new utility module `packages/graph-view/src/utils/elk-layout.ts` adds ELK-specific types and two public helpers: `repairElkInput` and `applyElkLayout`. It centralizes input validation, auto-correction, and layout execution for the graph view.

Key changes

- **Imports** added at lines 1-3: `GraphIssue`, `loadElk`, and `NODE_WIDTH/HEIGHT`.
- **New interfaces** defined at lines 5-34, 16-21, 22-27, 39-43, 214-216, 218-221.
- **`repairElkInput`** (lines 56-213) performs:
 - Fills missing node dimensions with `DEFAULT_NODE_WIDTH/HEIGHT`.
 - Deduplicates node IDs per parent.
 - Removes orphan children and edges.
 - Detects and strips containment cycles.
 - Emits `GraphIssue` objects and may throw when `opts.strict` is true.
- **`applyElkLayout`** (lines 223-243) loads ELK asynchronously, runs layout on the repaired input, and returns an `ElkLayoutResult`. On failure it returns a minimal positioned graph and a fatal issue unless `opts.strict` is true.

Impact

- Guarantees ELK receives well-formed input, reducing runtime layout errors.
- The `issues` array provides diagnostics for downstream consumers.
- Centralizes ELK logic; callers now import a single module instead of scattered helpers.
- No existing API is altered; the module is additive.

Risks & follow-ups

- `loadElk` must resolve correctly in all environments; a missing bundle could fail silently.
- `strict` flag behavior: errors propagate only when `opts.strict` is true.
- Default dimensions (`NODE_WIDTH/HEIGHT`) must stay in sync with the layout module to avoid visual inconsistencies.
- `applyElkLayout` should handle large graphs without stack overflows during recursive traversal.